

Hands-on workshop component

- Choose a dataset (one of your own, or one of the examples we provide)
 - Use examples and code provided throughout the slides
 - <https://github.com/veg/selection-tutorial/tree/main/data>
- Run several different selection analyses on this file
- Use HyPhy Vision to compile results and visualizations
- Think about your results, do they make sense?

Preliminaries

- Please confirm access to **HyPhy**: <http://hyphy.org/download/>
 - <https://youtu.be/fgNrPbOTpxE> (How to install HyPhy)
 - You can do a datamonkey.org based tutorial, but if you have Linux or OSX, you can also do a command line tutorial for more features.
 - General user questions and feedback: <https://github.com/veg/hyphy/issues>
- **Datamonkey** web-app: <http://www.datamonkey.org>
 - YouTube example videos (channel HyPhy vision)
 - <https://www.youtube.com/channel/UCIgRnbJjbOWhshe5ThhaWGW/videos> (Tutorials)
- Example datasets at <https://github.com/veg/selection-tutorial/>
 - Test datasets and practical instructions: www.hyphy.org (search for “Detect Selection”)

Important links

- hyphy.org (software main website - installation and tutorials)
- datamonkey.org (for browser-based analysis)
- <https://github.com/veg/selection-tutorial/> (additional information)
- <https://github.com/veg/selection-tutorial/tree/main/data> (precompiled datasets which can be used during the hands-on workshop)
- <https://www.datamonkey.org/citations> (Relevant literature on the methods implemented in HyPhy)

Assignment: Detecting selection pressures in viral genomes using HyPhy

Objective:

To understand how to use Hyphy for detecting selection pressures on viral genomes and to interpret the results in the context of viral evolution and pathogenesis.

Preparation

Ensure you have access to a computer with Hyphy installed.

Review the slides presented today on the basics of molecular evolution and selection analyses and decide which analysis best fits your hypothesis

Each student or group will choose a specific viral gene dataset (e.g., from HIV, Influenza, SARS-CoV-2) available from databases like GenBank or specific viral genome repositories. I recommend - <https://www.bv-brc.org/>

Data retrieval

Download the nucleotide sequences of their assigned viral gene.

Align the sequences using tools like MACSE

It is important to generate codon-aware alignments, otherwise the selection analyses will fail.

Hyphy Analysis

- Students will go through the following steps in Hyphy
 - **Input Preparation:** Import the aligned sequences into Hyphy.
 - **Model Selection:** Select an appropriate substitution model.
 - **Selection Analysis:** Run selection analysis using methods such as SLAC (Single Likelihood Ancestor Counting), FEL (Fixed Effects Likelihood), and MEME (Mixed Effects Model of Evolution).

Results Interpretation

Students will interpret the results of their analysis, focusing on:

- Identifying sites under positive or negative selection.
- Understanding the biological significance of these sites in the context of viral evolution.
- Comparing the results from different methods (SLAC, FEL, MEME) and discussing any discrepancies.

Report

- Each student or group should prepare a report that includes
 - A brief introduction to the viral gene and its significance
 - A methodology section outlining the steps taken in Hyphy.
 - Results with tables and/or graphs showing selected sites.
 - A discussion on the implications of the findings, including any potential functional or clinical relevance

Presentation

Students present their findings to the class, highlighting key points from their analysis and discussing any challenges they faced during the assignment.

Evaluation Criteria

- Accuracy and completeness of the data retrieval and alignment procedure
- Correct application and execution of Hyphy selection analyses.
- Depth of interpretation and biological insights drawn from the results.
- Clarity and coherence of the written report and oral presentation.